Berk Sezer

Undergraduate Researcher in Robotics, Vision, and Applied and Theoretical Machine Learning
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Publications

Same Actions, Different Context: Investigating Appropriateness of Robot Actions Across Environments, Embodiments, and Explanations
Submitted to the IEEE International Conference on Robotics and Automation (ICRA 2027, Main Conference, Under Review).
Fethiye Irmak Doğan*, **Berk Sezer***, Nikhil Churamani, Yiming Luo, Hatice Gunes (*Equal contribution / Co-first authors)

An Impossibility Result for ETF Target Geometry, and an Equinorm Remedy for Imbalanced Analytic Continual Learning
Accepted to the NeurIPS 2026 Workshop on Symmetry and Geometry in Neural Representations (NeurReps, Extended Abstracts Track).
Berk Sezer

Gaze4HRI: Zero-shot Benchmarking Gaze Estimation Neural-Networks for Human-Robot Interaction
Accepted to the 20th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2026, Main Conference).
Berk Sezer, Ali Görkem Küçük, Erol Şahin, Sinan Kalkan | Project Page

Education

Middle East Technical University (METU) — Ankara, Turkey
B.Sc. in Computer Science (4th Year), ***CGPA: 3.89/4.00,*** — *Oct 2023 - Present*
Ranked 7th/144 (Top 5%) in my class.

Advanced Coursework Highlight: **Graduate-level Deep Learning (Grade: 4.0/4.0)**
Implemented the training pipeline for a SOTA Video Blink Estimation network (DE-ViViT) from scratch (using PyTorch) for a term project reproducing papers lacking public code, ultimately surpassing the baseline performance reported in the original paper. Code

Research & Engineering Experience

University of Cambridge, Department of Computer Science and Technology — Cambridge, UK
Undergrad Researcher, Affective Intelligence and Robotics (AFAR) Lab — *July – September 2026*

Completed a full-time internship where I conducted research under the supervision of Prof. Hatice Gunes, initially exploring Continual and Federated Learning architectures, but later focusing on Cross-Domain Generalisation. Co-led the end-to-end benchmarking of a social-appropriateness of robotic actions dataset across CNN and VLM-LoRA models, encompassing pipeline implementation, experimental validation, and manuscript preparation resulting in a co-first-author ICRA 2027 submission.

METU ROMER (METU Robotics Center) — Ankara, Turkey
Undergrad Researcher & AI–Robotics Engineer — *Sep 2024 – Present*

Home Robotics Project (Beko R&D Partnership) – *AI–Robotics Engineer* — *(Part-Time)*
The Home Robotics Project aims to develop assistive robots for domestic environments. I worked under the direction of Prof. Erol Şahin, Prof. Sinan Kalkan, and Prof. Gökberk Cinbis, and have stepped back to focus on my ICRA 2027 and CVPR 2027 papers and the school term.

- **VLA Models:** Trained state-of-the-art Vision–Language–Action (VLA) models in both real-world and simulated settings.
- **Data Pipeline:** Led the architecture and implementation of our large-scale data collection pipeline to train VLA models.

USTA Human-Robot Interaction Project – *Undergrad Researcher*

The USTA Project advances Human–Robot Interaction (HRI) to enable collaborative robotic manipulators (cobots) to operate more naturally alongside humans. My work focuses on vision-based perception, specifically gaze estimation for HRI.

- **Gaze Estimation:** As first author of the IEEE FG 2026 main conference paper ($\sim 35\%$ acceptance rate), *"Gaze4HRI: Zero-shot Benchmarking Gaze Estimation Neural-Networks for Human-Robot Interaction"*, I led data collection, code implementation, experiment design, and manuscript writing.
- **Blink Estimation:** Finalizing the manuscript for CVPR 2027 on a novel blink estimation framework that I led. Contributed the core of the framework by applying temporal-order-based pre-training to enhance blink detection accuracy, which on its own substantially improves the majority of existing SOTA models as well as our own new model. On MPEblink 2.0, it beats prior SOTA on the main metric, Blink-AP, by over 80%.

METU Department of Computer Engineering Ankara, Turkey
Senior Project Member, MIMIC-Sound (team of 5, advisor: Prof. Sinan Kalkan) *Sep 2026 – Present*

MIMIC-Sound is an AI tutor that shows people how to physically change a sound they produce, instead of only scoring it. Project Page

- **Idea & Prototype:** Conceived the project and built a working prototype to address a gap in modern AI: machines can sing, speak and imitate any voice, but people still need clear vocal instructions. It adds articulatory motion as a new modality for mimicry, inferring it from audio alone, turning the gap to the target into a plain instruction built from sounds the learner already makes, and playing the corrected move on a physical vocal-tract synthesizer.

Industry Experience

Altima Sistem ve Yazılım Teknolojileri Ltd. Ankara, Turkey
Full-Stack Engineer *Aug 2022 – Jun 2023 (Full-Time)*

Developed and deployed a full-stack LLM + RAG application giving legally valid, citation-linked answers from legislative texts for the Union of Turkish Survey Engineers (HKMO), and engineered the core backend of randevuyla.com.

Technical Skills

Robotics & Data: ROS 2, sensor fusion, Vision–Language–Action (VLA) models, large-scale data collection pipelines.

Computer Vision & ML: Deep learning, self-supervised pre-training, continual learning, real-time 3D perception (RGB–D), face and gesture detection, gaze estimation.

Speech & Audio: Phone recognition, articulatory inversion, formant analysis, vocal-tract synthesis.

AI Assistants: VLM/LLM assistants with Retrieval-Augmented Generation (RAG) and structured outputs.

Programming: Python, C, C++, PHP, JavaScript/TypeScript.

DevOps & Deployment: Unix environments, HPC Servers, Docker.

Web Systems: Laravel, PostgreSQL, MySQL, Redis, MongoDB, Vue.js, HTML, CSS, TypeScript.

References

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Please feel free to contact any of the individuals listed above for inquiries. (symbols link to Google Scholar profiles.)